



ELEMENTARY MATH (CALCULUS)



EVALUATING LIMITS ALGEBRAICALLY

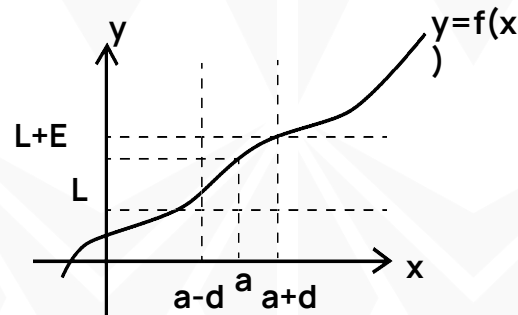
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LEARNING OBJECTIVES

At the end of this lesson, you should be able to:

- define the concept of limit with examples,
- evaluate limits algebraically using various methods like direct substitution, factoring and rationalization,
- determine limits involving indeterminate forms and apply L'Hôpital's rule,
- know the different properties of limit,
- Apply the concept of limit to the real world.

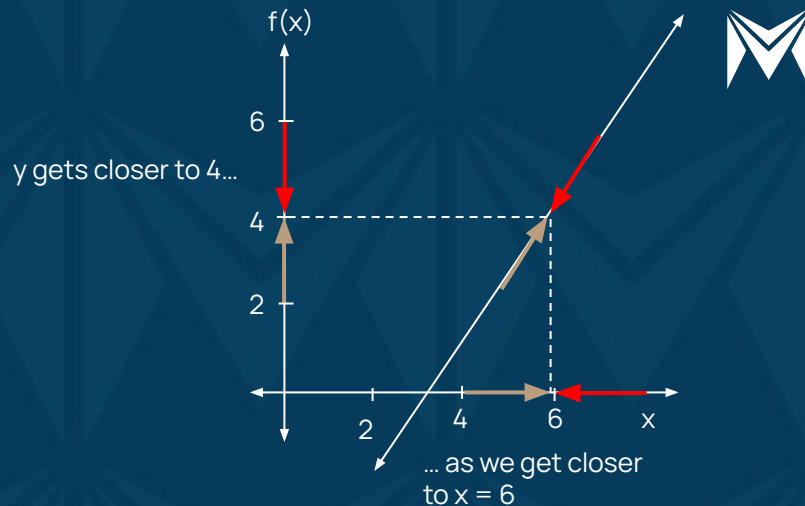




INTRODUCTION

A limit is a fundamental concept in calculus that describes the behavior of a function as its input (typically denoted as x) gets closer and closer to a specific point (usually denoted as c).

It represents the value that the function (y) approaches as x approaches c .





DIRECT SUBSTITUTION

This is the simplest method and applies when substituting the value of c directly into the function results in a well-defined expression.

Question: Evaluate $\lim_{x \rightarrow 3} x^2 + 2x - 3$

Solution: Substitute $x = 3$ into the expression:

$$\begin{aligned} &= (3)^2 + 2(3) - 3 \\ &= 9 + 6 - 3 \\ &= 15 - 3 \\ &= 12 \end{aligned}$$



DIRECT SUBSTITUTION

Question: Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 3}{x^2 - 6}$

Solution: Substitute $x = 2$ into the expression:

$$\begin{aligned} &= \frac{2^3 - 3}{2^2 - 6} \\ &= \frac{8 - 3}{4 - 6} = \frac{5}{-2} \\ &= -2.5 \text{ or } -2\frac{1}{2} \end{aligned}$$



FACRORIZATION METHOD

When direct substitution results in an indeterminate form, factoring and simplification can help.

Question: Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$

Solution: You notice how if we use direct substitution it will result to an indeterminate form. i.e. $\frac{2^2 - 4}{2 - 2} = \frac{0}{0}$

Now, let's factor the numerator: $(x - 2)(x + 2)$

so that, $\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)}$

Thus, cancel the common factor to get

$$\begin{aligned}\lim_{x \rightarrow 2} (x + 2) \\ &= (2 + 2) \\ &= 4\end{aligned}$$



FACTORIZATION METHOD

Question: Evaluate $\lim_{x \rightarrow 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2}$

Solution: This is a complex fraction. We multiply the numerator and denominator by the common denominator i.e. $2x$,

$$\lim_{x \rightarrow 2} \frac{2 - x}{2x(x - 2)}$$

Then, we factor out -1 , to get

$$\lim_{x \rightarrow 2} \frac{-1(x - 2)}{2x(x - 2)} = \lim_{x \rightarrow 2} \frac{-1}{2x} = \frac{-1}{2(2)} = \frac{-1}{4}$$



ELEMENTARY MATH (CALCULUS)

TASK

Find the limit of the function $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$





RATIONALIZATION

Useful when dealing with limits involving square roots or complex fractions.

Question: Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9}-3}{x}$

Solution: Rationalize the numerator by multiplying both the numerator and denominator by the conjugate of the numerator:

$$\lim_{x \rightarrow 0} \frac{(\sqrt{x+9}-3)}{x} \cdot \frac{(\sqrt{x+9}+3)}{(\sqrt{x+9}+3)} = \lim_{x \rightarrow 0} \frac{(x+9)-9}{x(\sqrt{x+9}+3)} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+9}+3)}$$

Since x is common, cancel it out so that we have:

$$\lim_{x \rightarrow 0} \frac{1}{(\sqrt{x+9}+3)} = \frac{1}{\sqrt{0+9}+3} = \frac{1}{\sqrt{9}+3}$$

$$= \frac{1}{3+3} = \frac{1}{6}$$



RATIONALIZATION

Question: Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4}$

Solution:

$$\begin{aligned} & \lim_{x \rightarrow 4} \frac{\sqrt{x}-2}{x-4} \cdot \frac{\sqrt{x}+2}{\sqrt{x}+2} \\ &= \lim_{x \rightarrow 4} \frac{(x-4)}{(x-4)(\sqrt{x}+2)} \\ &= \lim_{x \rightarrow 4} \frac{1}{\sqrt{x}+2} \\ &= \frac{1}{\sqrt{4}+2} = \frac{1}{2+2} = \frac{1}{4} \end{aligned}$$



ELEMENTARY MATH (CALCULUS)

TASK

Question: Evaluate $\lim_{x \rightarrow 4} \frac{\frac{1}{\sqrt{x}} - \frac{1}{2}}{x - 4}$





L'HOPITAL'S RULE

L'Hôpital's Rule essentially allows you to replace the original function with its derivatives in situations where direct substitution results in an indeterminate form.

Question: Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

Solution: Given $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$,

and applying L'Hôpital's rule, such that

$$f(x) = \sin x, g(x) = x$$

$$f'(x) = \cos x, g'(x) = 1$$

we have $\lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$, to be $\lim_{x \rightarrow 0} \frac{\cos x}{1}$

$$= \frac{\cos(0)}{1} = \frac{1}{1} = 1$$



L'HOPITAL'S RULE

Question: Evaluate $\lim_{x \rightarrow \infty} \frac{x^2}{e^x}$

Solution:

$$\lim_{x \rightarrow \infty} \frac{2x}{e^x}$$

Since direct substitution results to an indeterminate form, so we differentiate again,

$$\lim_{x \rightarrow \infty} \frac{2}{e^x} = \frac{2}{e^{\infty}} = \frac{2}{\infty} = 0$$



PROPERTIES OF A LIMIT

Properties of limits are fundamental rules that help simplify the evaluation of limits and provide insights into the behavior of functions as they approach certain values. Here are some of the key properties of limits:

- Limit of a Sum
- Limit of a Difference
- Limit of a Constant
- Constant Multiple Rule
- Limit of a Product
- Limit of a Quotient
- Limit of a Power
- Limit of a Composite Function
- Squeeze Theorem
- Limit of a Reciprocal



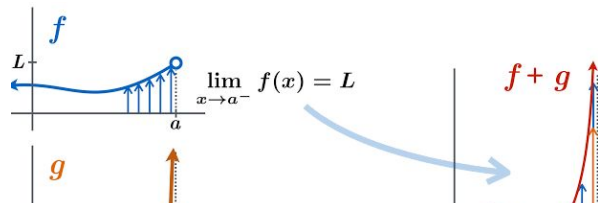
LIMIT OF A SUM

$$\lim_{x \rightarrow c} [f(x) + g(x)] = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$$

In other words, the limit of a sum is the sum of the limits.

For example,

$$\begin{aligned}\lim_{x \rightarrow 2} (3x + 4x) &= \lim_{x \rightarrow 2} 3x + \lim_{x \rightarrow 2} 4x \\ &= 3(2) + 4(2) \\ &= 6 + 8 \\ &= 14\end{aligned}$$





LIMIT OF A DIFFERENCE

$$\lim_{x \rightarrow c} [f(x) - g(x)] = \lim_{x \rightarrow c} f(x) - \lim_{x \rightarrow c} g(x)$$

In other words, the limit of a difference is the difference of the limits.

For example,

$$\begin{aligned}\lim_{x \rightarrow -3} (7x - 2x) &= \lim_{x \rightarrow -3} 7x - \lim_{x \rightarrow -3} 2x \\ &= 7(-3) - 2(-3) \\ &= -21 + 6 \\ &= -15\end{aligned}$$



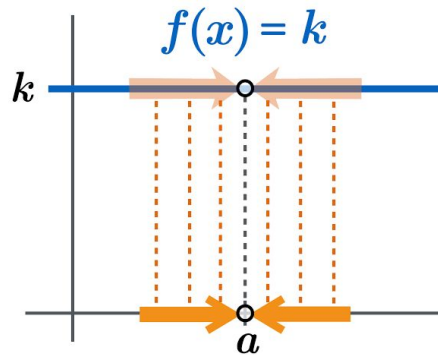
LIMIT OF A CONSTANT

$$\lim_{x \rightarrow c} k = k$$

The limit of a constant is the constant itself.

For example,

$$\lim_{x \rightarrow 1} 5 = 5$$





CONSTANT MULTIPLE RULE

$$\lim_{x \rightarrow c} kf(x) = k \cdot \lim_{x \rightarrow c} f(x)$$

A constant multiple can be factored out of the limit.

For example,

If $\lim_{x \rightarrow c} f(x) = 4$, what is $\lim_{x \rightarrow c} 3f(x)$?

Solution:

Recall from the constant multiple rule that

$$\begin{aligned}\lim_{x \rightarrow c} 3f(x) &= 3 \cdot \lim_{x \rightarrow c} f(x) \\ &= 3(4) \\ &= 12\end{aligned}$$



LIMIT OF A PRODUCT

$$\lim_{x \rightarrow c} [f(x) \cdot g(x)] = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x)$$

The limit of a product is the product of the limits.

For example,

If $\lim_{x \rightarrow 2} (2x) = 4$, and $\lim_{x \rightarrow 2} (3x) = 6$.

Then,

$$\begin{aligned} \lim_{x \rightarrow 2} [(2x) \cdot (3x)] &= \lim_{x \rightarrow 2} (2x) \cdot \lim_{x \rightarrow 2} (3x) \\ &= 4 \cdot 6 \\ &= 24 \end{aligned}$$





LIMIT OF A QUOTIENT

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$$

provided $\lim_{x \rightarrow c} g(x) \neq 0$.

The limit of a quotient is the quotient of the limits as long as the denominator's limit is not zero.

For example,

$$\lim_{x \rightarrow 4} \frac{6x^2}{5x} = \frac{\lim_{x \rightarrow 4} 6x^2}{\lim_{x \rightarrow 4} 5x} = \frac{6(4)^2}{5(4)} = \frac{96}{20} = 4.8$$





LIMIT OF A POWER

$$\lim_{x \rightarrow c} [f(x)^n] = [\lim_{x \rightarrow c} f(x)]^n$$

The limit of a power is the power of the limit.

For example,

$$\begin{aligned}\lim_{x \rightarrow 0} (x + 4)^3 &= [\lim_{x \rightarrow 0} (x + 4)]^3 \\ &= (0 + 4)^3 \\ &= 4^3 \\ &= 64\end{aligned}$$



LIMIT OF A COMPOSITE FUNCTION

$$\lim_{x \rightarrow c} f(g(x)) = f \lim_{x \rightarrow c} g(x)$$

Provided that $\lim_{x \rightarrow c} g(x)$ is in the domain of $f(x)$.

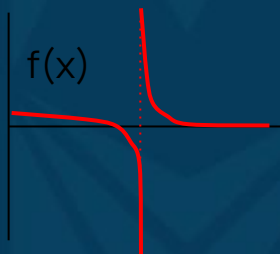
The limit of a composite function is the same as applying the limit to the inner function.

Example: Find $\lim_{x \rightarrow 1} f(g(x))$ if $f(x) = x^2$ and $g(x) = x - 3$.

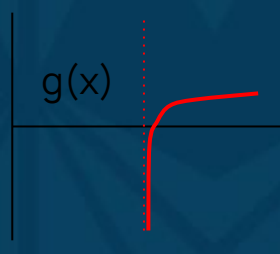
Solution: Recall that $\lim_{x \rightarrow 1} f(g(x)) = f \lim_{x \rightarrow 1} g(x)$

$$f \lim_{x \rightarrow 1} (x - 3) = f(1 - 3) = f(-2)$$

$$\begin{aligned} \text{Therefore, } f(x) = x^2 &= f(-2) \\ &= (-2)^2 = 4 \end{aligned}$$



$$\lim g(f(x))$$



$$\lim g(f(x))$$





SQUEEZE THEOREM (SANDWICH THEOREM)

If $f(x) \leq g(x) \leq h(x)$ for all x in some interval containing c except possibly at c itself.

$$\lim_{x \rightarrow c} f(x) = \lim_{x \rightarrow c} h(x) = L, \text{ then } \lim_{x \rightarrow c} g(x) = L$$

This theorem helps determine the limit of a function trapped between two other functions with known limits.

For example,

If $5 - x^3 \leq f(x) \leq 5 + x^3$ what is

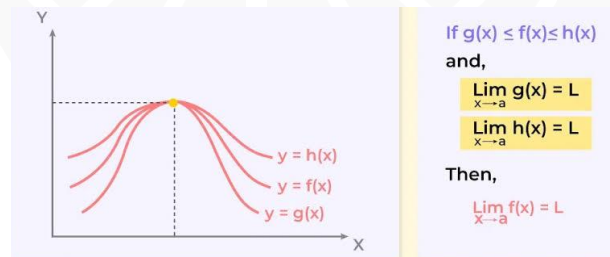
$\lim_{x \rightarrow 0} f(x)$?

$$\lim_{x \rightarrow 0} (5 - x^3) \leq f(x) \leq \lim_{x \rightarrow 0} (5 + x^3)$$

From the Squeeze theorem,

$$(5 - 0^3) \leq f(x) \leq (5 + 0^3)$$
$$5 \leq f(x) \leq 5$$

Therefore, $\lim_{x \rightarrow 0} f(x) = 5$.





LIMIT OF A RECIPROCAL

$$\lim_{x \rightarrow c} \frac{1}{f(x)} = \frac{1}{\lim_{x \rightarrow c} f(x)}$$

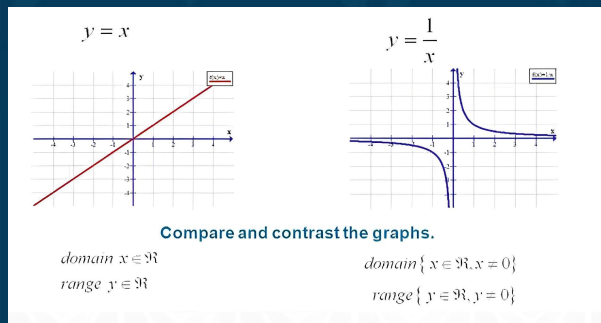
provided $\lim_{x \rightarrow c} f(x) \neq 0$.

The limit of a reciprocal is the reciprocal of the limit, as long as the limit of the original function is not zero.

Example: Given $\lim_{x \rightarrow 6} \left(\frac{1}{x^2} \right) = \frac{1}{\lim_{x \rightarrow 6} (x^2)}$, prove that the left-hand side is equal to the right-hand side.

$$\text{Solution: } \left(\frac{1}{6^2} \right) = \frac{1}{(6^2)} \Rightarrow \frac{1}{36} = \frac{1}{36}$$

Therefore, the L.H.S=R.H.S





APPLICATIONS OF

LIMIT

Limits play a fundamental role in various fields, allowing researchers and professionals to model, analyze, and solve real-world problems in physics, engineering, economics, and computer science. Here's how limits are used in each of these fields:

1. Control Systems: Engineers use limits to analyze the stability of control systems. By evaluating limits, they can determine whether a system's response converges or diverges, which is crucial for designing stable and reliable control systems for various applications.

2. Risk Assessment: Limits help financial analysts assess risk by modeling the behavior of financial instruments and portfolios under different market conditions.



APPLICATIONS OF LIMIT

3. Microeconomics: Limits are used to analyze marginal concepts, such as marginal cost, marginal revenue, and marginal utility. These concepts help economists make decisions about resource allocation and pricing.

4. Macroeconomics: Economists apply limits to study economic growth, inflation, and interest rates. Limits help model how economies evolve over time and how they react to various factors.





SUMMARY

1. Limit describes the behavior of a function as its input (typically denoted as x) gets closer to a specific point (usually denoted as c).

2. It's notation is $\lim_{x \rightarrow c} f(x)$

3. $\lim_{x \rightarrow \infty} f(x)$ represents the limit approaching x from the right

while $\lim_{x \rightarrow -\infty} f(x)$ is used when the limit approaches x from the left on the graph.

4. From the various properties of limits, we can evaluate limits algebraically using various methods.



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FURTHER READING

"Calculus" by James Stewart.

"Calculus: Early Transcendentals" by Howard Anton, Irl Bivens, and Stephen Davis.

"Calculus" by Ron Larson and Bruce H. Edwards.





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THANK YOU